

## **Task 2 - AI ML Technology Research: What are different technologies and trends and jobs**

### **AI/ML Technology Research**

#### **❖ Introduction -**

Artificial Intelligence (AI) and Machine Learning (ML) are technologies that allow computers to “think” and “learn” from data. They are used everywhere today—from recommending movies on Netflix to detecting fraud in banks. This report explores the different technologies, current trends, and jobs in AI/ML to understand how they shape industries and careers.

#### **❖ AI/ML Technologies -**

AI/ML relies on a variety of tools and frameworks that help engineers and scientists build smart systems.

- ☐ Machine Learning Frameworks: Tools like *TensorFlow*, *PyTorch*, *Scikit-learn* are used to build and train models. *Example:* A developer uses TensorFlow to create a model that predicts house prices.
- ☐ Deep Learning Tools: Libraries like *Keras* and *Hugging Face Transformers* help build advanced models such as neural networks. *Example:* Hugging Face is used to build chatbots that understand human language.
- ☐ Data Handling Tools: Tools like *Pandas*, *NumPy*, *Spark* help clean, organize, and process large datasets. *Example:* Analysts use Pandas to prepare sales data before training a model.
- ☐ Cloud Platforms: Services like *AWS AI/ML*, *Google Cloud AI*, *Microsoft Azure ML* provide ready-made infrastructure for training and deploying models. *Example:* A company deploys its fraud detection model on AWS to handle millions of transactions daily.
- ☐ Specialized Areas:

- Computer Vision: AI that recognizes images (used in facial recognition, medical imaging).
- Natural Language Processing (NLP): AI that understands text and speech (used in chatbots, translation apps).
- Reinforcement Learning: AI that learns by trial and error (used in robotics and gaming).

### ❖ AI/ML Trends -

AI/ML is evolving rapidly, and new trends are shaping industries worldwide.

- ☐ Generative AI: Tools like *ChatGPT* and *DALL·E* create text, images, and even music. *Example*: Businesses use ChatGPT to automate customer support.
- ☐ Edge AI: Running AI models directly on devices like smartphones or IoT gadgets. *Example*: Smart cameras detect suspicious activity without needing cloud servers.
- ☐ AI in Healthcare: AI helps doctors analyze medical images and predict diseases. *Example*: AI systems detect early signs of cancer in X-rays.
- ☐ AI in Finance: Used for fraud detection, credit scoring, and algorithmic trading. *Example*: Banks use AI to flag unusual transactions instantly.
- ☐ Automation & Robotics: AI powers self-driving cars, warehouse robots, and delivery drones. *Example*: Amazon uses AI robots to move packages in warehouses.
- ☐ Ethical AI: Focus on fairness, transparency, and reducing bias in models. *Example*: Companies test AI hiring tools to ensure they don't favor one gender or group unfairly.

## ❖ AI/ML Jobs -

AI/ML has created a wide range of career opportunities.

- ☐ AI Engineer: Builds and deploys AI systems into real-world applications. *Example*: An AI Engineer develops a chatbot for an e-commerce site.
- ☐ Machine Learning Engineer: Designs and trains ML models. *Example*: An ML Engineer at Spotify builds models to recommend songs.
- ☐ Data Scientist: Analyzes data and creates predictive models. *Example*: A Data Scientist at Amazon studies shopping patterns to improve recommendations.
- ☐ Research Scientist: Works on cutting-edge AI innovations. *Example*: A Research Scientist develops new algorithms for autonomous vehicles.
- ☐ Business Analyst (AI Projects): Connects business needs with AI solutions. *Example*: A Business Analyst ensures an AI system meets customer service goals.
- ☐ AI Product Manager: Oversees AI-driven product development. *Example*: An AI Product Manager manages the rollout of a voice assistant app.

## ❖ Conclusion -

AI/ML technologies are transforming industries by automating tasks, predicting outcomes, and creating new opportunities. Current trends like *Generative AI*, *Edge AI*, and *Ethical AI* show how fast the field is growing. For professionals, AI/ML offers diverse career paths - from engineers and scientists to analysts and product managers. As businesses continue to adopt AI, the demand for skilled talent will only increase.